

HARSHDEEP SINGH

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Summary

Backend-focused Software Engineering intern building AI-integrated backend systems in Python/FastAPI and Node.js/Express. Experienced in modular service design, REST API architecture, and independent ownership of production backend code.

Experience

Software Engineering Intern — AI Automation — Zeber

June 2026 – Present

- Serve as the sole engineer building and maintaining AI-powered backend systems for automation workflows, owning technical decisions from initial architecture through deployment.
- Built modular backend services and REST APIs in Python and FastAPI, separating routes, business logic, and data-handling into independently testable components.
- Debug and iterate on backend code independently, tracing issues through logs and service boundaries to identify root causes before shipping fixes.
- Incorporate engineering feedback into architecture revisions, refactoring service boundaries and API contracts as requirements evolve.
- Maintain the codebase through structured error handling, configuration management, and disciplined version control across iterative feature releases.

Projects

AI Voice Agent

Python, FastAPI, Google Gemini, Whisper STT, Edge TTS, WebRTC VAD

- Built a conversational AI backend structured as independently maintainable services for speech recognition, LLM orchestration, conversation memory, and speech synthesis.
- Structured the execution pipeline to coordinate transcription, LLM inference, state tracking, and voice synthesis across each stage of a conversational turn.
- Developed a dispatcher-based tool execution layer that decouples intent recognition from the tools it invokes, allowing new tools to be added without modifying core conversation logic.
- Exposed conversational logic through REST APIs, isolating backend reasoning from client-side interaction.

GatePulse — API Gateway (In Progress)

Node.js, Express.js, JavaScript

- Structuring a production-oriented API gateway with distinct routing, controller, service, and validation layers to keep concerns separated as the system grows.
- Building API key lifecycle management — creation, validation, and revocation — through dedicated service and routing layers with schema-based request validation.
- Applying middleware-based validation to reject malformed requests before they reach business logic.

Technical Skills

Languages: Python, JavaScript, C++, Java, SQL

Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, Middleware

AI & LLM: Google Gemini, Whisper STT, Edge TTS, WebRTC VAD, Tool Calling, Conversation Memory

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, Postman, VS Code

Education

Bharati Vidyapeeth College of Engineering (BVCOE), GGSIPU

Expected 2028

B.Tech in Information Technology — CGPA: 8.47/10

Leadership

National Cadet Corps (NCC) — Corporal — Promoted from Lance Corporal to Corporal in recognition of leadership, discipline, and responsibility.

Content Head — NCC MAIT — Led planning and execution of official digital content while coordinating publishing workflows.